

Incuvate v. Penumbra

Potential expert disciplines and candidates for a five-patent dispute concerning aspiration-pressure monitoring in Penumbra's THUNDERBOLT thrombectomy platform.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
August 19, 2026	N.D. California	4:26-cv-08658	Patent infringement

Prepared August 25, 2026

No candidate contacted. Availability, interest, compensation and conflicts not checked.

The principal infringement questions concern how THUNDERBOLT measures and interprets aspiration pressure.

The asserted claims appear to require more than pressure sensing in the abstract. The likely disputes include whether the accused system detects a specified series of vacuum-pressure changes, determines baseline and peak pressure values, compares a pressure difference with a stored threshold, and produces the claimed alerts.

CAPTION	Incuvate, LLC v. Penumbra, Inc.
FORUM / DOCKET	United States District Court for the Northern District of California / 4:26-cv-08658
FILED / TYPE	2026-08-19 / Patent infringement
ACCUSED PLATFORM	Penumbra THUNDERBOLT computer-assisted vacuum thrombectomy platform, including Engine, THUNDERBOLT tubing/module, and RED reperfusion catheters.
PUBLIC DOCKET	Open docket

ANTICIPATED EXPERT DISCIPLINES

Counsel may benefit from complementary testimony: an aspiration-thrombectomy or biofluid-mechanics expert for the physical and clinical operation, and a pressure-sensor, embedded-systems or signal-processing expert for the detection logic. A practicing neurointerventionalist may also be useful for method-of-use and instruction evidence.

LIKELY RESEARCH PURCHASER

Timothy B. Yoo - Partner and Chair, Intellectual Property Litigation Practice Group, Bird Marella. He is the plaintiff filing attorney on the public docket and chairs the firm's IP litigation practice.

tyoo@birdmarella.com - publicly listed on the [official Bird Marella biography](#); not inferred from an email pattern.

Commercial context: Penumbra reported 2025 thrombectomy revenue of \$947.9 million and company-wide revenue of \$1.4037 billion in its 2025 Form 10-K. THUNDERBOLT Aspiration Tubing is listed under FDA 510(k) K250690.

Asserted claims identified in the public complaint analysis: [US 9,248,221, claim 5](#) | [US 9,895,473, claim 1](#) | [US 10,603,415, claim 1](#) | [US 11,678,896, claim 1](#) | [US 12,539,127, claim 11](#)

Questions likely to require expert testimony

01 Does THUNDERBOLT detect a claimed pattern comprising multiple increases and decreases in vacuum pressure, rather than ordinary pump ripple, noise, or an arbitrary fluctuation?

Potential evidence: Source code, signal definitions, sampling and filtering specifications, logged pressure traces, and controlled bench testing under partial and complete occlusion.

Disciplines: biomedical fluid mechanics / signal processing / embedded sensing

02 How does the accused system establish baseline and peak vacuum pressure, calculate their difference, and compare that value with a stored threshold?

Potential evidence: Algorithm and firmware review, memory parameters, validation protocols, and repeatable waveform analysis.

Disciplines: controls / signal processing / sensor systems

03 Are the pressure sensor, tubing, module, catheter, and vacuum source arranged in the fluid-communication and inline/removable configurations required by the asserted claims?

Potential evidence: Teardown, schematics, fluid-path inspection, sensor characterization, and claim-construction analysis.

Disciplines: medical-device engineering / MEMS and pressure sensors / catheter design

04 Do the system's visual and audible outputs correspond to distinct claimed states - fluctuation detected and fluctuation no longer detected - and are they triggered by the claimed logic?

Potential evidence: IFUs and training materials, event logs, source code, human-factors documentation, and synchronized bench video/data.

Disciplines: embedded systems / human factors / clinical workflow

Questions likely to require expert testimony

05 Which catheter, clot, blood-analog, tubing-compliance, pump, and operator variables materially change the pressure signature and alert performance?

Potential evidence: A validated test matrix spanning clot analogs, vessel geometries, catheter sizes, flow states, noise, latency, and occlusion scenarios.

Disciplines: thrombectomy / vascular biomechanics / experimental fluid mechanics

06 Would public instructions, demonstrations, or training direct clinicians to perform every step of the asserted method claims?

Potential evidence: IFUs, clinical training, launch materials, recorded demonstrations, and clinician testimony mapped element by element.

Disciplines: neurointerventional practice / medical-device instructions

07 What did a skilled person understand about aspiration-pressure monitoring, clot detection, thresholds, and aspiration modulation at the claimed priority date?

Potential evidence: Patent-family history, contemporaneous papers, earlier devices, prior-art systems, and a defensible POSITA definition.

Disciplines: patent technical analysis / thrombectomy history / biofluid mechanics

08 What technical information was conveyed in the alleged 2016 NDA-era meeting and the 2026 notices, and does THUNDERBOLT engineering reflect that information?

Potential evidence: Meeting records, technical decks, design timelines, version control, inventor and engineer testimony, and an independent provenance comparison.

Disciplines: medical-device development / technical chronology

Candidates 1-2 for further diligence

Ranked for plaintiff-side research priority based on technical fit, public work, testimony and visible independence-screen burden. This is not a retention recommendation.

01 Matthew J. Gounis, PhD

Professor of Radiology and Director, New England Center for Stroke Research, UMass Chan Medical School

TECHNICAL FIT

Experimental fluid mechanics, preclinical stroke-device evaluation, aspiration thrombectomy, emboli and clot-device interaction.

RELEVANT WORK

Leads a translational stroke-research center and has published extensively on thrombectomy devices, in-vitro and in-vivo models, and procedural mechanics.

PUBLIC LITIGATION EXPERIENCE

No public patent-litigation testimony was located in the bounded review.

POINTS FOR FURTHER DILIGENCE

Primarily an engineering and preclinical researcher rather than a practicing operator; industry collaborations and funding would require detailed conflicts diligence.

[Source 1](#) | [Source 2](#)

[Email expert](#) | Matthew.Gounis@umassmed.edu | [email source](#)

02 Michael Chen, MD

Professor of Neurology, Neurosurgery and Radiology, Rush University Medical Center

TECHNICAL FIT

Practicing neurointerventionalist with aspiration-thrombectomy research and device-study leadership.

RELEVANT WORK

National principal investigator on an aspiration stroke study; published on endovascular devices and thrombectomy practice.

PUBLIC LITIGATION EXPERIENCE

Submitted a declaration and was deposed for RapidPulse in Penumbra v. RapidPulse PTAB proceedings concerning aspiration thrombectomy technology.

POINTS FOR FURTHER DILIGENCE

Prior work adverse to Penumbra creates heightened confidentiality, conflict and repeat-engagement diligence; clinical expertise does not replace an engineering signal-analysis expert.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Email expert](#) | Michael_Chen@rush.edu | [email source](#)

Candidates 3-4 for further diligence

03 Ender A. Finol, PhD

Zachry Endowed Chair and Chair of Mechanical, Aerospace and Industrial Engineering, University of Texas at San Antonio

TECHNICAL FIT

Vascular biomechanics, biofluid mechanics, computational fluid dynamics, and device design.

RELEVANT WORK

Leads research in cardiovascular biomechanics and computational/experimental modeling; Fellow of ASME, AHA, AIMBE and BMES.

PUBLIC LITIGATION EXPERIENCE

Submitted expert opinions and deposition testimony for RapidPulse in a Penumbra PTAB proceeding involving aspiration-thrombectomy fluid mechanics and POSITA issues.

POINTS FOR FURTHER DILIGENCE

Prior RapidPulse engagement creates conflict and confidentiality diligence; his broader cardiovascular CFD work is not specific to THUNDERBOLT's production algorithm.

[Source 1](#) | [Source 2](#)

[Email expert](#) | ender.finol@utsa.edu | [email source](#)

04 Nabil Simaan, PhD

Professor of Mechanical Engineering, Vanderbilt University

TECHNICAL FIT

Surgical robotics, assistive perception, pressure sensing, signal classification, and catheter-thrombus contact detection.

RELEVANT WORK

Coauthored a 2024 study using proximal pressure sensing and vacuum excitation to detect catheter-thrombus contact with machine learning; named inventor on related assistive-perception patent work.

PUBLIC LITIGATION EXPERIENCE

No public litigation testimony was located in the bounded review.

POINTS FOR FURTHER DILIGENCE

Active inventorship in closely overlapping pressure-sensing technology creates meaningful IP, commercial-interest and neutrality diligence.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Email expert](#) | nabil.simaan@vanderbilt.edu | [email source](#)

Candidates 5-6 for further diligence

05 Michael T. Froehler, MD, PhD

Associate Professor of Neurology, Neurosurgery and Radiology and Director, Cerebrovascular Program, Vanderbilt University Medical Center

TECHNICAL FIT

Neurointerventional practice, stroke thrombectomy, and translational fluid-mechanics research.

RELEVANT WORK

Coauthored the 2025 review 'Fluid mechanics of aspiration thrombectomy' and directs a clinical cerebrovascular program.

PUBLIC LITIGATION EXPERIENCE

No relevant public patent-litigation testimony was located in the bounded review.

POINTS FOR FURTHER DILIGENCE

Clinical and trial relationships require conflicts diligence; a separate engineering expert would likely be needed for firmware and signal-processing opinions.

[Source 1](#) | [Source 2](#)

[Email expert](#) | m.froehler@vanderbilt.edu | [email source](#)

06 Thomas W. Kenny, PhD

Richard W. Weiland Professor and Senior Associate Dean for Student Affairs, Stanford School of Engineering

TECHNICAL FIT

MEMS, sensors, microstructures, force and pressure measurement, and medical-device engineering.

RELEVANT WORK

Longstanding Stanford sensor researcher and inventor with extensive work on microfabricated sensing systems.

PUBLIC LITIGATION EXPERIENCE

Public CV materials identify more than ten expert matters, including medical devices and physiological sensors.

POINTS FOR FURTHER DILIGENCE

Not a thrombectomy clinician and no aspiration-specific publication was located; he would need a clinical or biofluids counterpart.

[Source 1](#) | [Source 2](#)

[Email expert](#) | kenny@cdr.stanford.edu | [email source](#)

Candidates 7-8 for further diligence

07 Bryan Good, PhD

Assistant Professor of Biomedical Engineering, University of Tennessee, Knoxville

TECHNICAL FIT

Cardiovascular biomechanics, experimental and computational thrombectomy models, and cyclic aspiration.

RELEVANT WORK

Leads a cardiovascular biomechanics laboratory and coauthored the 2025 review 'Fluid mechanics of aspiration thrombectomy.'

PUBLIC LITIGATION EXPERIENCE

No public litigation testimony was located in the bounded review.

POINTS FOR FURTHER DILIGENCE

Relatively junior and no testifying record was found; academic models would need validation against production-device data.

[Source 1](#) | [Source 2](#)

[Email expert](#) | bgood6@utk.edu | [email source](#)

08 Rohan V. Chitale, MD

Associate Professor of Neurological Surgery, Mechanical Engineering and Radiology, Vanderbilt University Medical Center

TECHNICAL FIT

Neuroendovascular surgery, thrombectomy workflow, catheter-thrombus contact, and pressure-sensing research.

RELEVANT WORK

Coauthored the vacuum-excitation catheter-thrombus detection study and is named on related assistive-perception patent work.

PUBLIC LITIGATION EXPERIENCE

No public litigation testimony was located in the bounded review.

POINTS FOR FURTHER DILIGENCE

Closely overlapping inventorship creates IP and commercial-interest diligence; clinical expertise may need a signal-processing complement.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Email expert](#) | rohan.v.chitale@vumc.org | [email source](#)

Candidates 9-10 for further diligence

09 Robert D. Howe, PhD

Abbott and James Lawrence Professor of Engineering, Harvard University

TECHNICAL FIT

Medical robotics, catheter force sensing, biomechanics, controls, haptics, and sensor-guided intervention.

RELEVANT WORK

Leads Harvard's BioRobotics Laboratory and has published on robotic catheter systems combining image guidance, force sensing, and closed-loop control.

PUBLIC LITIGATION EXPERIENCE

Public PTAB records show deposition testimony in Intuitive Surgical v. Ethicon concerning surgical-robotics technology.

POINTS FOR FURTHER DILIGENCE

His catheter work is cardiac and robotic rather than aspiration thrombectomy; he would need a thrombectomy or biofluids counterpart and a conflict review of industry collaborations.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Email expert](#) | howe@seas.harvard.edu | [email source](#)

10 Robert J. Webster III, PhD

Richard A. Schroeder Professor of Mechanical Engineering and Senior Associate Provost for Commercialization and Technology Transfer, Vanderbilt University

TECHNICAL FIT

Surgical robotics, mechanically and sensor-guided intervention, device design, controls, and technology translation.

RELEVANT WORK

Heads Vanderbilt's Medical Engineering and Discovery Laboratory; Vanderbilt reports more than 75 patents and applications and two startup companies arising from his work.

PUBLIC LITIGATION EXPERIENCE

No public expert-testimony record was located in the bounded review.

POINTS FOR FURTHER DILIGENCE

No aspiration-thrombectomy publication was located; extensive patenting, startups, licensing and industry work require careful commercial-interest and conflicts diligence.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Email expert](#) | Robert.webster@vanderbilt.edu | [email source](#)

Preliminary candidates; no conflicts or availability determination.

- Run party, affiliate, counsel, inventor, funder and technology conflicts.
- Confirm testimony history, exclusions, compensation, publications and deposition performance.
- Investigate patents, licensing, grants, consulting and commercial interests.
- Confirm role fit, availability and interest only after counsel authorizes outreach.

Initial independence screen: Known Penumbra-side experts in closely related PTAB matters were excluded from the ranked plaintiff-side slate. Candidate-specific weaknesses remain diligence items, not conflict conclusions.

CORE MATTER SOURCES

Justia public docket

<https://dockets.justia.com/docket/california/candce/4%3A2026cv08658/476629>

Ex Parte public complaint analysis

<https://ai-lab.exparte.com/case/dct/cand/4%3A26-cv-08658/doc/analysis/1>

Bloomberg Law report

<https://news.bloomberglaw.com/california-brief/incuvate-says-penumbra-infringed-blood-clot-detection-systems>

Bird Marella official biography for Timothy B. Yoo

<https://www.birdmarella.com/attorneys/timothy-b-yoo/>

FDA 510(k) K250690

<https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?ID=K250690>

Penumbra 2025 Form 10-K

<https://www.sec.gov/Archives/edgar/data/1321732/000132173226000007/pen-20251231.htm>

Penumbra THUNDERBOLT launch announcement

<https://www.penumbrainc.com/penumbras-thunderbolt-receives-fda-clearance-bringing-computer-assisted-vacuum-thrombectomy-technology-to-stroke/>

LIMITATIONS

- No candidate was contacted.
- Availability, interest, compensation and conflicts were not checked.
- Public-source litigation-history searches are incomplete and must be supplemented before engagement.
- The complaint analysis is secondary; counsel should replace it with the filed complaint and exhibits from PACER or the case file before relying on paragraph-level allegations.
- No legal conclusion on infringement, validity, damages, willfulness or inducement is offered.

The docket and attorney biography were verified directly. The technical issue map also uses a public paragraph- and claim-level complaint analysis; its direct page was not live at final verification and the public docket did not expose the complaint PDF. Replace the secondary analysis with the filed complaint and exhibits before relying on allegation details.